

Encoding Multimodal Scenes in a Virtual City: How the Multifaceted Self Shapes Naturalistic Episodic Memory

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Figure 1: Example of naturalistic multimodal daily life events. Left: A child jumps into a puddle of water in front of his father, who gets annoyed that his child is getting dirty. Right: A street performance where a woman dances while a man plays the violin to accompany her.

ABSTRACT

The self-reference effect (SRE) suggests that encoding new material in episodic memory (EM) is enhanced when information is closely linked to the Self. However, traditional approaches to the SRE have neglected the interplay between multiple facets of the Self and naturalistic, multisensory EM contexts. The present study investigates the respective and joint contributions of the minimal and narrative Selves to the SRE during the encoding of naturalistic, multisensory daily events in a virtual city, experienced while embodied in a first-person avatar. Preliminary findings suggest that the narrative Self plays a predominant role in enhancing EM encoding under ecological conditions.

Index terms: Episodic Memory, Minimal Self, Embodiment, Narrative Self, Self-Reference Effect, Virtual Reality.

1 INTRODUCTION

The Self encompasses mental processes that provide individuals with a sense of uniqueness, coherence, individuality, and continuity over time [1]. It has two main interconnected facets. On the one hand, the minimal Self refers to the pre-reflective awareness of being an embodied agent experiencing and acting within the multisensory world. It encompasses the senses of body ownership, self-location, first-person perspective, and agency, and relies on the multimodal integration of coherent and synchronous multisensory bodily signals [2]. On the other hand, the narrative Self refers to the reflective construct of one's self-concept, and encompasses

conceptual knowledge about the Self (e.g., personality traits, values, personal goals), self-esteem, and self-image [1]. The Self has been shown, both theoretically and in laboratory contexts, to modulate cognition, particularly episodic memory (EM), which refers to the memory of the Self's past experiences. It corresponds to the recollection of unique, personally experienced events, situated in time and space, enriched with phenomenological details [4]. The Self acts as a processing bias that determines how and what information is encoded and retrieved [1]: according to the self-reference effect (SRE), processing information closely related to the Self constitutes the most effective strategy for remembering new material [5].

However, traditional SRE studies have primarily focused on the minimal or narrative Selves separately, and relied on the encoding of simplistic items, which fail to capture the richness of EM and its connections with the Self in daily life. Immersive virtual reality (IVR) appeared as a valuable technology for enabling more naturalistic and comprehensive investigations of the interplay between the multifaceted Self and EM in everyday contexts [6,7].

The study aimed to reassess the SRE on EM through an integrated and naturalistic approach, using IVR to examine the individual and joint contributions of the minimal and narrative Selves to the SRE on the EM of naturalistic, multimodal daily events.

2 MATERIAL AND METHODS

2.1 Participants

A power analysis conducted using the *R simr* package indicated that a sample size of 60 participants would achieve 0.91 power for the model. Recruitment is ongoing; to date, forty French-speaking healthy young adults have been enrolled, with 85% of women, a mean age of 20.6 ± 2.3 years, and 13.2 ± 1.7 years of education.

2.2 VR equipment, virtual environment, and stimuli

Immersion in the virtual environment was achieved using the HTC Vive Pro Eye kit (Taoyuan City, Taiwan: HTC Corporation), which included full-body tracking with six tracking points for avatar

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embodiment, and eye-tracking. The virtual environment was created using Unity 2019.2.5f1. It consisted of a reproduction of a city, comprising 40 naturalistic daily life events, which featured characters, animals, or objects in specific situations, with visual and auditory animations (Fig. 1).

2.3 Experimental conditions

The study used a two-way within-subjects design. The virtual city was divided into two parts, each containing 20 events. Participants navigated one part with a synchronous avatar and the other with an asynchronous avatar (minimal Self: strong vs. weak embodiment conditions). Within each city part, participants rated the significance of half of the events for themselves, and the significance of the other half for another person (narrative Self: self vs. other conditions). ‘Other’ was defined as the French President, Emmanuel Macron, to ensure the referent was universally known among participants. Two versions of the virtual environment were created, with identical events but reversed assignments of Self and Other conditions for each event (e.g., if Event 1 was assigned to Self in version X, then it was assigned to Other in version Y). The versions of the environment, the order of avatar synchrony conditions, and the order of visited city parts were all counterbalanced across participants.

2.4 General procedure

The full-body ownership illusion was induced in a generic gender-matched first-person avatar before each visit to the city, in a virtual room equipped with a mirror. It relied on synchronous visuomotor stimulation (strong embodiment) or asynchronous stimulation with a 650 to 1000 ms visuomotor delay (weak embodiment).

Following embodiment induction, participants navigated one part of the city while embodied in either a synchronous or asynchronous avatar. They were instructed to photograph each encountered event and rate its significance for either Self or Other, by responding to the question: “To what extent does this event have a particular significance for [You/Macron], i.e., to what extent does it reflect [your/his] personality, values, tastes, projects, or motivations?”. This navigation process also enabled an incidental encoding of the events in EM.

After completing both navigations and complementary measures, participants underwent a verbal free recall test to assess their EM of the naturalistic events (IVR-EM test). They were asked to indicate all the events they remembered (what), where they took place in relation to themselves (where) and at which moment of the navigation (when), in which navigation it was (source – navigation), whether the significance of the event was assessed in relation to themselves or the other person (source – referent), and event-specific objective and phenomenological details. A binding score was calculated as a measure of associative memory, and sense of recollection was assessed using the Remember/Know paradigm. The IVR-EM test was administered a first time around 40 minutes after encoding, and a second time with a one-month delay to study long-term memory consolidation.

Participants also completed self-reported questionnaires and neuropsychological tests to collect additional data to refine the analyses and explore correlations (e.g., assessments of minimal Self, narrative Self, sense of embodiment and presence in IVR, cybersickness, emotional state, cognitive load, verbal fluency, executive functioning, and autobiographical EM).

3 PRELIMINARY RESULTS

A generalized linear model with a binomial link function examined the effects of *Avatar Synchrony* and *Referent* on event recall (1: event recalled; 0: not recalled) (Fig. 2). The results revealed a significant main effect of *Referent* ($\beta = 0.33$, $p = .036$), indicating

that events related to Self were more likely to be recalled than events related to Other. No significant effects were found for *Avatar Synchrony* or the interaction between *Avatar Synchrony* and *Referent*. A linear mixed model examining the effects of *Avatar Synchrony* and *Referent* on sense of recollection also revealed a significant main effect of *Referent* ($\beta = 4.90$, $p = .008$), but no significant effects of *Avatar Synchrony* or interaction (Fig. 2). Although non-significant, smaller differences between Self and Other conditions were observed in the synchronous compared to asynchronous avatar condition for event recall, number of phenomenological details, and sense of recollection.

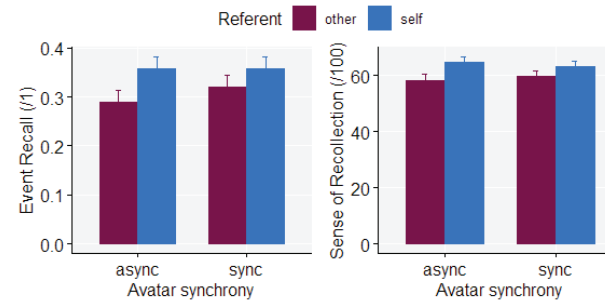


Figure 2: Event Recall and Sense of Recollection (N=40).

4 CONCLUSION

The study investigated how the minimal Self, rooted in bodily experiences, and the narrative Self, grounded in higher-order conceptual representations of the Self, shaped the encoding of naturalistic, multisensory events into EM, providing valuable insights into how humans perceive, interpret, and remember their multimodal experiences. Preliminary results suggest that the narrative Self plays a predominant role in enhancing EM encoding under ecological conditions. Furthermore, the minimal Self may contribute to enhance EM in contexts where the narrative Self is not engaged.

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REFERENCES

- [1] Prebble, S. C., Addis, D. R., & Tippett, L. J. (2013). Autobiographical Memory and Sense of Self. *Psychological Bulletin*, 139(4), 815–840. <https://doi.org/10.1037/a0030146>
- [2] Blanke, O. (2012). Multisensory Brain Mechanisms of Bodily Self-Consciousness. *Nature Reviews Neuroscience*, 13(8), 556–571. <https://doi.org/10.1038/nrn3292>
- [3] Tulving, E. (2002). Episodic Memory: From Mind to Brain. *Annual Review of Psychology*, 53, 1–25. <https://doi.org/10.1146/annurev.psych.53.100901.135114>
- [4] Klein, S. B. (2012). Self, Memory, and the Self-Reference Effect: An Examination of Conceptual and Methodological Issues. *Personality and Social Psychology Review*, 16(3), 283–300. <https://doi.org/10.1177/1088868311434214>
- [5] Piolino, P. (2022). La réalité virtuelle pour une approche écologique de la mémoire épisodique: Évaluation et prise en charge en neuropsychologie. In *Neuropsychologie Clinique et Technologies* (pp. 243–276). De Boeck Supérieur, Louvain-la-Neuve, Belgium.
- [6] Penaud, S., Yeh, D., Gaston-Bellegarde, A., & Piolino, P. (2023). The Role of Bodily Self-Consciousness in Episodic Memory of Naturalistic Events: An Immersive Virtual Reality Study. *Scientific Reports*, 13(1), 17013. <https://doi.org/10.1038/s41598-023-43823-2>